

EDUCATION

Doctor of Philosophy - PhD in Artificial Intelligence <i>Ecole Polytechnique</i>	2026 - 2029
Master of Science: MVA Mathématiques, Vision, Apprentissage <i>Institut Polytechnique de Paris - Ens Paris-Saclay - GPA: 4.0</i>	2023 - 2024
Master 1 in Applied Mathematics <i>Institut Polytechnique de Paris - Telecom Paris - GPA: 4.0</i>	2022 - 2023
Engineer's Degree in Data Science and Artificial Intelligence <i>INP ENSEEIHT</i>	2020 - Present

TECHNICAL EXPERIENCE

AI Research Scientist - Sanofi <i>Generative AI for Drug Development</i>	Jan 2026 — Present
- Designing reasoning-driven multi-agent LLM systems for PK/PD and mechanistic model construction, combining tool-augmented LLMs with simulation, verification, and structured scientific workflows. - Conducting benchmarking and evaluation of AI methods for pharmacology, assessing robustness, interpretability, and alignment of generative and reasoning models with domain-specific constraints. - Developing diffusion models and flow-matching approaches for molecular graph generation, focusing on validity, controllability, and scalability for drug discovery applications. - Exploring reinforcement learning and preference-based optimization to enforce chemical and pharmacological validity, guiding generative models toward feasible and high-quality molecular candidates.	
AI Engineer - Cartes Bancaires <i>Real-Time Fraud Detection – IAGO Model</i>	Nov 2024 — Nov 2025
- Implemented a new version of the model responsible of detecting fraud in billions of transactions per month using advanced machine learning techniques achieving a 21% increase in overall performance. - Developed a model monitoring pipeline and a Causal Concept-based Explanations approach through black-box distillation to interpret and explain the decisions made by previous versions of the real-time IAGO model. - Conducted research on transforming current datasets into time-dynamic graphs to enhance the fraud detection framework and improve real-time anomaly identification.	

PROJECTS & COMPETITIONS

Self-Distilling Conversational Agent with RLHF <i></i>	2025
- Designed a personalized chatbot fine-tuned with Reinforcement Learning from Human Feedback (RLHF), integrating self-distillation to compress large models while preserving alignment quality. - Implemented a human-in-the-loop training loop to iteratively improve responses, highlighting skills in efficient LLM training, preference alignment, and inference-time optimization.	
Investigating Flow Matching and Diffusion Models <i></i>	2025
- Implemented conditional flow matching and rectified flow methods, benchmarking on CIFAR-10 and synthetic datasets to study training stability and generalization. - Explored retraining on synthetic data as a form of model distillation, linking empirical observations to optimal transport theory and efficient sampling.	
Multimodal Retrieval with Contrastive Representation Learning <i></i>	2024
- Built a multimodal retrieval system combining text encoders with structured representations (molecules, products), trained via contrastive learning. - Demonstrated the capacity of cross-modal alignment techniques for semantic search and knowledge grounding, extending beyond domain-specific retrieval.	
Reinforcement Learning for Forecasting and Control <i></i>	2023
- Developed a hybrid pipeline for retail demand forecasting, combining statistical time-series models with RL-based adaptive policy control. - Showcased the ability to link representation learning on sequential data with decision-making under uncertainty, bridging theory and large-scale applied problems.	

ACHIEVEMENTS

Two times Tunisian National Mathematical Olympiad Champion
Two times Tunisian Representative in the Mathematics World Cup